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A framework of trajectory design and optimization for the hypersonic gliding vehicle

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ABSTRACT

Trajectory of Hypersonic Gliding Vehicles (HGV) designed and optimized generally is an especial path with actual task requirements, which mostly provide feasible flight information for the design of guidance and control system. This paper mainly presents a framework of trajectory design and optimization for HGV, and differences between design and optimization results are discussed. In the first part of the framework, the trajectory is designed mainly by the programming angle commands. Specifically, the pitch angle as the control variable in the boost phase, and the angle of attack (AOA) as the control variable in the glide phase are considered. Then, establishing a trajectory optimization model, which has coincident initial and terminal conditions with the trajectory design part if it's reasonable, and additional constraints. Applying a Legendre-Gauss-Radau (LGR) orthogonal collocation method for solving the problem. In this part, Firstly, low density collocation points are set in the algorithm to find the feasible trajectory, and then further increase the density of collocation points to optimize to get more information about the state and control of the trajectory points. Finally, several initial parameters of the glide phase are discussed, which have a great influence on the flight performance. Numerical simulation indicates that the effectiveness of the proposed framework for the real task scenarios is amazing.

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1. Introduction

HGV like CAV, HTV, HTV-2, X-37B, AHW and YU-71 have been drawing the attention of researchers and scientists. During the past few years, HGV have obtained great development around the world, which reflects the strength of countries' military power [1–3]. Normally, HGV is launched by the rocket propulsion or released on a certain height by other vehicle and accelerate to climb later on, then HGV starts to glide unpowered outside the atmosphere by utilizing the aerodynamics when it climbs to a certain altitude or mission field, eventually attacking the target or landing. Dai et al. [4] proposed a morphing waverider based on the variable Mach number theory, and its aerodynamic performance was investigated comprehensively. The variable Mach number theory was first proposed by Li et al. [5]. Typically, the flight phase can be divided into three parts: boost phase, gliding phase and reentry phase [6]. The trajectories of the boost and gliding phases are the most critical part of the flight distance, as well as its control [7,8], and the trajectories of the boost and gliding phases are investigated in detail in this paper.

Trajectory design and optimization is aim to find an optimal path meeting various environmental and flight state constraints under the basic mission conditions, and it is based on its parametric design [9]. It is a feasibility design that trajectory design is to seek and verify whether the generated path is reasonable and valid and meets basic task requirements. The focus of the trajectory optimization is to find a desired path matching the desired performance index. From extensive studies, there are two kinds of trajectory optimization methods, namely analytical and numerical methods, and the analytical method can be generalized into two categories: indirect and direct methods [10]. The indirect methods are based on the classical variational method or the Pontryagin minimum (maximum) value principle, which transforms the optimal control problem into a Hamilton two-point boundary value problem [11]. Although the optimality can be guaranteed, a good initial guess of the adjoint variables is required, which have no explicit physical meaning in most cases. The direct method has risen with the development of the computer technology, and it includes direct shooting method, collocation method, differential inclusion method, Pseudospectra method and so on, which do not require an initial value to perform the optimization. For the numerical op-

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timization method, the sequential quadratic programming method can well solve the smooth nonlinear programming problem and occupies a dominant position in the field of the trajectory optimization [12–16].

Because of the strong nonlinear coupling among aerodynamic forces, flight commands, constraints, trajectory states and powerful performance index of HGV, the gliding trajectory optimization problem is extremely challenging. Fahroo and Ross [17] generated the optimal trajectory by using a Chebyshev Pseudospectral method, which is one of the direct methods. Although the method can solve based optimal problem without explicit initial guess, it belongs to a kind of local optimization algorithm, when it involves multiple extremum, it may fall into the local optimal solution. Murillo and Lu [18] generated fuel optimal ascent trajectories for the air-breathing launch vehicle. The problem was posed as a two-point boundary value problem and was solved using finite difference method for finding the optimum angle of attack control to achieve the required target states with minimum fuel consumption. Xie et al. [19] respectively designed longitudinal and lateral paths in consideration of multi-constraints. The longitudinal subplanner is designed to confirm the altitude profile and the magnitude of the bank angle. Grant and Braun [20] utilized the indirect method to solve the conceptual trajectory design of hypersonic missions, which although the optimal control can be obtained, requiring a precise initial guess of the adjoint variables. Liu and Shen [21] applied the second-order cone programming to hypersonic vehicles with high lift-to-drag ratio. The non-convexity of the problem is first addressed via successive linearization and relaxation techniques; then the trajectory is optimized using a primal-dual interior-point algorithm. Patterson et al. [22] developed an improved Gauss-pseudo-spectral method for solving multiple-phase optimal control problems, which has high precision compared to the traditional method. They also developed it as a MATLAB software as an application. Meanwhile, Chai et al. [23] applied the approach for space maneuver vehicles trajectory optimization, which confirmed its effectiveness. Forbes-Spyratos et al. [24] described the trajectory optimization of a three-stage launch vehicle, which has scramjet propulsion in the second stage. But either they only study the influence on trajectory optimization of the change of performance indexes and constraints, or improve the optimization methods, however, there are few literatures on the comprehensive description of trajectory design and optimization methods.

To discuss the difference between trajectory design and optimization, and explore the optimization problem of an actual and highly constrained trajectory, this paper presents a framework of trajectory design and optimization and comparing the results between design and optimization, in a sense, which can provide experience guide for the accuracy of trajectory design problem. Firstly, uncontrolled trajectory design with the pitch angle as the control variable in the ascending phase, considering AOA as the control variable in the gliding phase according to a specific task requirement. The control laws of the pitch angle and AOA are given. Secondly, the trajectory optimization is attempted using a LGR orthogonal collocation method. The state variables and control variables are discretized on a series of Gauss points. At the same time, these points are used as nodes to construct Lagrange interpolation polynomials to approximate state variables and control variables. The derivative of state variable to time is estimated by the derivative of global interpolation polynomials, which makes state equations an algebraic constraint. After the above transformation, the optimal control problem is converted to a nonlinear programming (NLP) problem, and SQP algorithm is utilized to solve it. The optimization objective is to reach the specified height with minimum time in the boost stage and obtain maximum gliding distance in the gliding stage where HGV always moves out of the atmosphere and satisfy the path constraints of dynamic pressure, overload and heat flux, as well as the control variables, terminal values of the state. Finally, simulation validates the practicability of proposed framework.

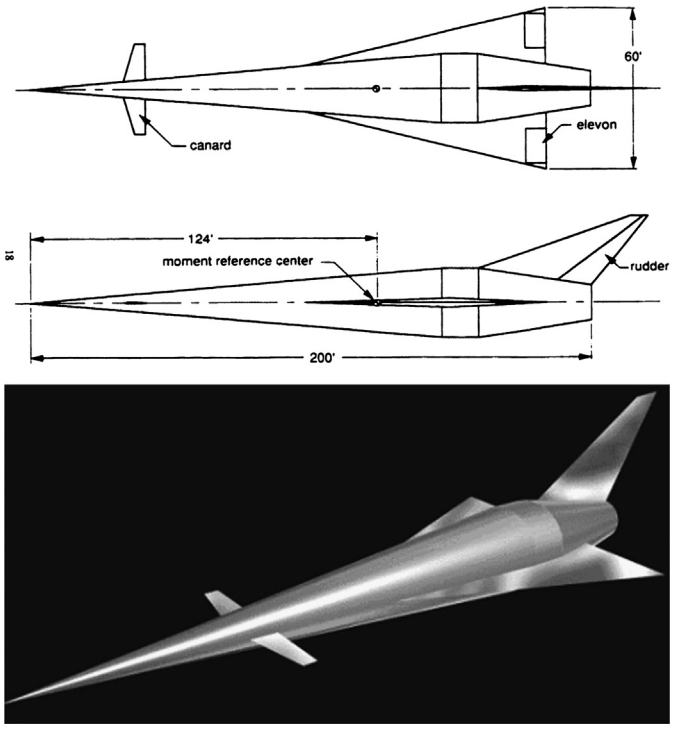


Fig. 1. Winged-cone configuration.

The sketch of the paper is as follows. Section 2 details the dynamic equations of HGV. In Section 3, the framework of trajectory design and optimization is proposed and giving the specific solving process, which optimization model is formulated. Section 4 carries out the simulation and Section 5 provide the conclusions.

2. Vehicle model

For simplicity, this paper is based on the following assumptions:

(I) Ignore the curvature of the earth, and the ground coordinate system is the inertial coordinate system.

(II) Instantaneous equilibrium hypothesis.

(III) Only longitudinal motion is considered.

The six-degree-of-freedom winged-cone hypersonic gliding vehicle model is selected to demonstrate the reliability of the framework. The vehicle configuration is shown in Fig. 1, which adopts the integrated structure of body and propulsion system and the vehicle consists of four control surfaces, right elevon, left elevon, rudder, and canard. The influence of rudder and canard will be ignored in the conditions of assumptions. The precise model parameters are expressed in Ref. [25].

2.1. Model dynamics equations

The longitudinal dynamics model can be described with the following equations:

$$\frac{dV}{dt} = \frac{T\cos\alpha - D}{m} - g\sin\gamma \tag{1}$$

$$\frac{d\gamma}{dt} = \frac{L + T\sin\alpha}{m} - \frac{g}{V}\cos\gamma \tag{2}$$

$$\frac{d\alpha}{dt} = q - \frac{d\gamma}{dt} \tag{3}$$

$$\frac{dx}{dt} = V\cos\gamma \tag{4}$$
$$\frac{dy}{dt} = V\sin\gamma \tag{5}$$
$$\frac{dm}{dt} = -m_c \tag{6}$$

Here, $g = g_0(\frac{R_e}{r})^2$ is the acceleration of gravity, g_0 is the acceleration of gravity on the ground and R_e is the radius of the earth, m_c represents the consumption of the fuel.

The moment of inertia expressions for the vehicle is given as [26]

$$I_{yy} = I(Ma) \tag{7}$$

2.2. Force and moment calculations

Calculations of forces and moments are presented in this subsection.

The lift (L) and drag forces (D) are descripted as follows:

$$L = \bar{q}SC_L \tag{8}$$
$$D = \bar{q}SC_D \tag{9}$$

where $\bar{q}$ is defined as $\bar{q} = \frac{1}{2}\rho V^2$ explained as dynamic pressure, and ρ represents the atmospheric density and varies with the height, which adopts an exponential form as that stated in Ref. [27]:

$$\rho = \rho_0 \exp\left(\frac{h_0 - h}{h_s}\right) \tag{10}$$

The terms C_L, C_D are lift and drag coefficients respectively.

In general, thrust is a function of Mach number, altitude, AOA and throttle opening that is written as follows in Ref. [28]

$$T = T(Ma, h, \alpha, \delta_p) \tag{11}$$

The aerodynamics coefficients are given as Refs. [25,26]. Applying the expression of supersonic and hypersonic because the speed of HGV is always more than sound velocity.

3. Trajectory optimization problem statement

Aircraft trajectory optimization problems usually have the following forms:

$$\begin{aligned} \min\ & J = \psi(X(\tau_f), t_f) \\ \text{s.t.}\ & \dot{X}(\tau) = \tfrac{t_f - t_0}{2} f(X(\tau), U(\tau), \tau; t_0, t_f) \\ & \vartheta(X(\tau_0), t_0) = 0 \\ & \phi(X(\tau_f), t_f) = 0 \\ & C_i(X(\tau), U(\tau), \tau; t_0, t_f) \le 0,\ i = 1, \ldots, n \end{aligned} \tag{12}$$

Here, $X(t) = [V(t)h(t)\ \gamma(t)\ l(t)\ m(t)]^T$ represents the state variants, $\vartheta(X(\tau_0), t_0) = 0$ is the initial value of the states, the terminal constraint conditions are $\phi(X(\tau_f), t_f) = 0$, and $C_i(\tau)$ refers to other constraints.

3.1. Objective function

During the boost stage, we hope that the HGV can climb rapidly in order to reach the expected altitude, velocity, flight path angle. Therefore, the optimization target in this stage is the terminal time, and the optimization performance index is

$$J = t_f \tag{13}$$

The objective of the glide stage is to maximize the transverse distance, and it is expressed by

$$J = -\int_{t_0}^{t_f} V\cos\gamma\, dt \tag{14}$$

3.2. Constraints

Because of the divided two-stage flight process, there are different optimization constraints in different phases according to the flight characteristics.

3.2.1. Path constraints

During the boost phase, generally, the variation of the heat flux density of the vehicle does not exceed its maximum limitation, thus just considering the dynamic pressure constraints:

$$q = \frac{1}{2}\rho V^2 \le q_{\max} \tag{15}$$

However, it must be considered that the influence of the heat flux density will change constantly in the process of gliding and maneuvering, so it is necessary to restrict it.

$$q = \frac{1}{2}\rho V^2 \le q_{\max} \tag{16}$$
$$\dot{q}_s = K_p \rho^{0.5} V^{3.08} \le \dot{q}_{s\max} \tag{17}$$

3.2.2. Control variant constraints

In the trajectory design part, the pitch angle is adopted in the boost stage in order to directly control the longitudinal attitude of the aircraft to change according to the expected flight state. On the contrary, the AOA is used as the control variant in both the boost and glide phases in the optimization design. The reason is that, on the one hand, the AOA is controlled so that there is no sudden change in the transition process from the boost phase to the glide phase. On the other hand, the AOA is also controlled to maintain flying with a large lift-drag ratio to enlarge the flight distance.

Therefore, during both stages, the AOA is restricted by

$$\alpha^i_{\min} \le \alpha \le \alpha^i_{\max}, \quad i = 1, 2 \tag{18}$$

3.2.3. Boundary constraints

During the boost phase, the initial and terminal constraints are given as follows,

$$V^1(t_0) = V^1_0 \qquad V^1(t_f) = V^1_f \tag{19}$$
$$h^1(t_0) = h^1_0 \qquad h^1(t_f) = h^1_f \tag{20}$$
$$\gamma^1(t_0) = \gamma^1_0 \qquad \gamma^1(t_f) = \gamma^1_f \tag{21}$$

During the glide phase, there are also similar expressions,

$$V^2(t_0) = V^1_f \qquad V^2(t_f) = V^2_f \tag{22}$$
$$h^2(t_0) = h^1_f \qquad h^2(t_f) = h^2_f \tag{23}$$
$$\gamma^2(t_0) = \gamma^1_f \qquad \gamma^2(t_f) = \gamma^2_f \tag{24}$$

4. A framework of trajectory design and optimization

In this subsection, a unified trajectory design and optimization framework for HGV is illustrated in Fig. 3.

Considering a general mission flight scenario. HGV is carried by a booster to a predetermined altitude, and then released. After that, the scramjet engine starts to ignite for climbing under the condition that the HGV has had a certain initial speed. This stage is called boost stage. The engine is shut down to enter the glid-

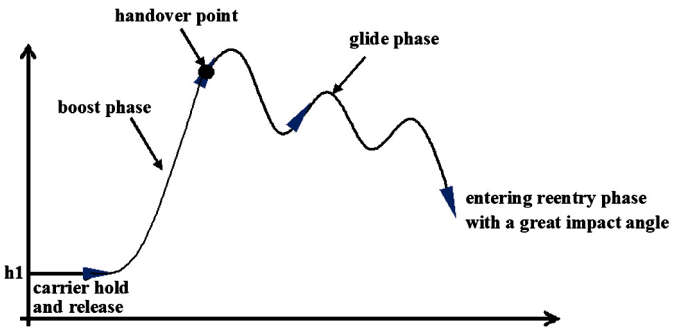


Fig. 2. General flight scenario.

ing flight state when the vehicle meets altitude conditions. Later on, the gliding mission is accomplished when the vehicle reaches reentry point, as shown in Fig. 2.

Step 1: According to the actual scene requirements, segment the trajectory, and the programming angle commands of each stage are designed to obtain the feasible trajectory scheme by applying interpolation method for solving the equation to calculate trajectory parameters.

Step 2: Check whether all parameters of the trajectory meet the design requirements. If the conditions are met, turn to the end. If not, establishing the trajectory optimization model. Select the performance optimization index and set relative constraints.

Step 3: The problem of HGV trajectory optimization is solved by a LGR orthogonal collocation method [2].

Step 4: Analyze the initial value of trajectory design part, which will be taken as the initial conditions and the final value as the constraint conditions if it's feasible. Otherwise, reset the initial and terminal values. Then, set the low-density collocation points to solve the optimization problem for feasible solution.

Step 5: Considering to take high-density collocation points to resolving the problem for reflecting more state and control information.

Step 6: Finally, the results for the trajectory optimization problem of HGV are output.

5. Results and discussion

5.1. Trajectory design parameters

Taking the winged-cone HGV as an example. The trajectory design simulation is carried out according to the mission flight scenario proposed in Section 4. The trajectory of the boost phase adopts the pitch angle control mode, and the specific expression form is as follows:

$$\theta = \begin{cases} \theta_0 & h \le h_1 \\ \theta_0 - \frac{C_1}{3(1+\exp(-(t-n)/m))}, & \text{else} \end{cases} \tag{25}$$

Herein, θ_0 is related to time, and its maximum value is 20°. C_1, n, m are all constants, and h_1 stand for the judgment conditions of the variation of the pitch angle, which are set to be $C_1 = \frac{pi}{4}$, $n = 200$, $m = 60$ and its function curve is shown in Fig. 4(a).

The trajectory of the glide phase adopts the AOA control mode, and its specific design form is as follows:

$$\alpha = \begin{cases} \alpha_0 & V_1 \le V \le V_0 \\ \frac{\alpha_{\max L/D} - \alpha_0}{V_2 - V_1}(V - V_1) + \alpha_0, & V_2 \le V \le V_1 \\ \alpha_{\max L/D} & V_f \le V \le V_2 \end{cases} \tag{26}$$

Here, V_0 and V_f represent the initial and terminal velocities of the gliding process respectively, V_1, V_2 are the speeds of the segment points, and they are considered as the design parameters. In this paper, $V_1 = 2500$ m/s, $V_2 = 1500$ m/s. α_0, $\alpha_{\max L/D}$ denote the maximum AOA constraint and the AOA of the maximum lift-to-drag ratio respectively, which are set to be $\alpha_0 = 10°$, $\alpha_{\max L/D} = 5°$ respectively. Its function curve is shown in Fig. 4(b).

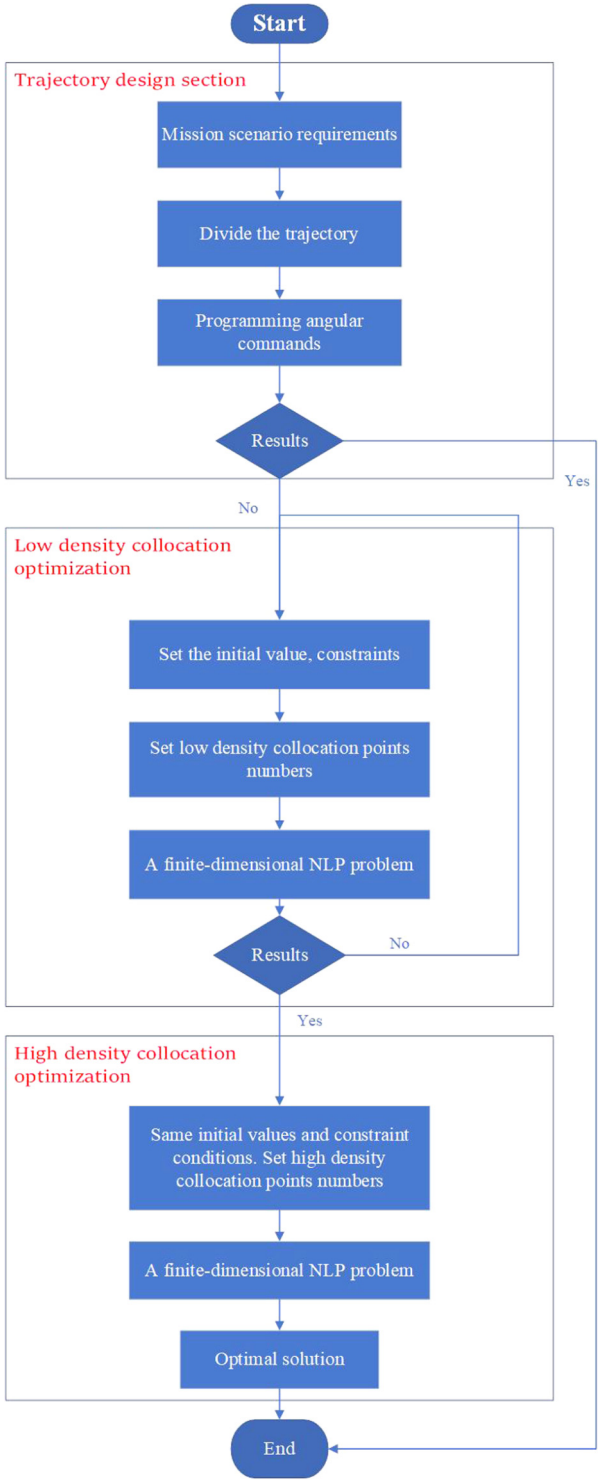


Fig. 3. A framework of trajectory design and optimization for HGV.

5.2. Comparison and discussion of the trajectory design and optimization results

The initial conditions of the boost and glide phases are as follows in Table 1.

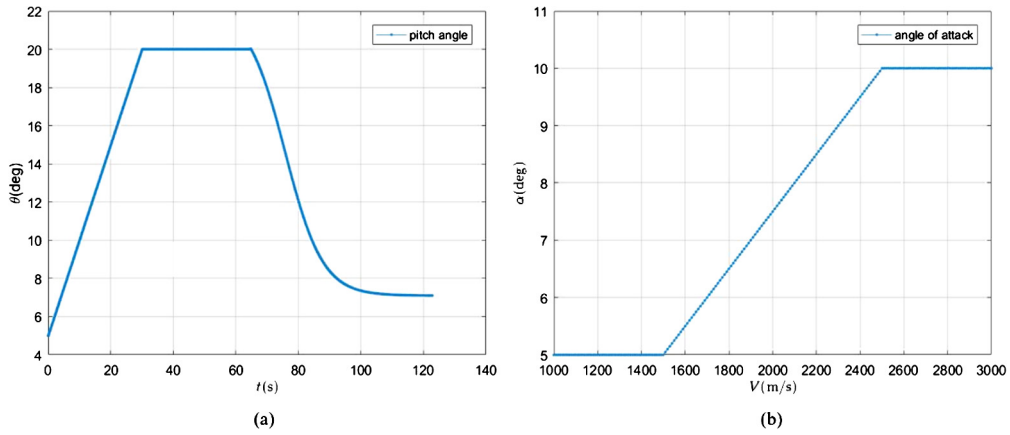


Fig. 4. Trajectory design parameters. (a) The profile of pitch angle. (b) The profile of AOA.

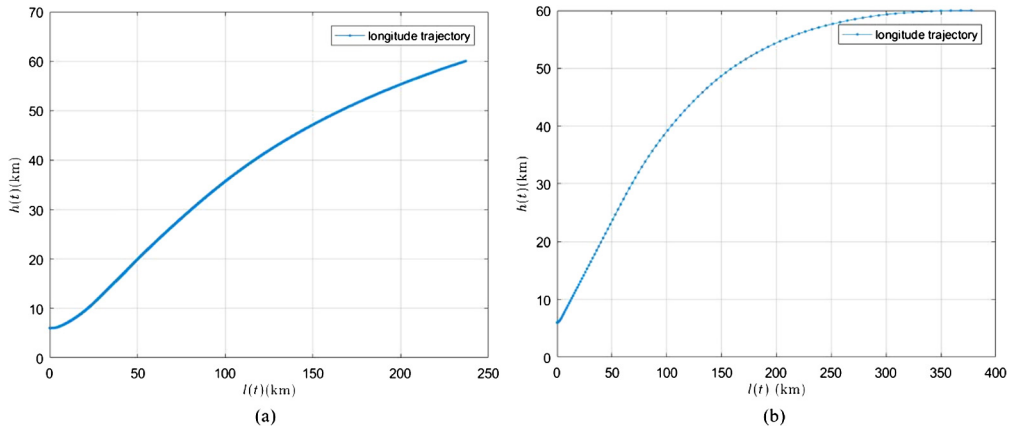


Fig. 5. The trajectory profile of the boost phase. (a) Designed trajectory profile. (b) Optimization trajectory profile.

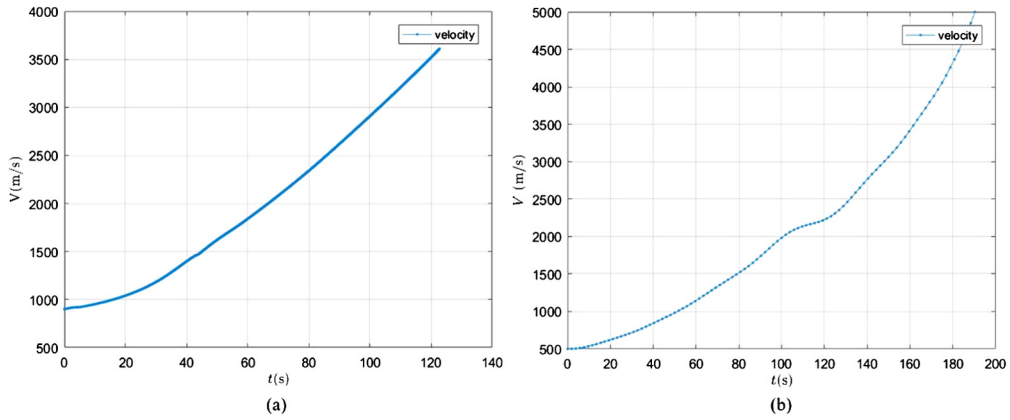


Fig. 6. The velocity profile of the boost phase. (a) Designed trajectory profile. (b) Optimized trajectory profile.

From the results of trajectory design, the dynamic pressure of the vehicle (Fig. 5(a)) is larger than the constraint condition before the 45 s. And the transition conditions of the boost phase and the glide phase (Fig. 6(a) and Fig. 7(a)) can't meet the requirements. Therefore, it is necessary to limit the speed and other parameters, and optimize the trajectory.

The optimization parameters and constraints are shown in Table 2.

Comparing the results both the design part and optimization part. In the boost phase of design part, flight altitude is selected as the judgment condition of the end. Its value is set $h = 60$ km. The total flight time of HGV is 122.8 s and the distance is 237.067 km. The optimization results indicate that the vehicle needs more time to reach the preset terminal conditions, which can be seen from Fig. 6(b) and Fig. 7(b). At the same time, as can be seen that the velocity of HGV in design part reaches 3610 m/s but don't meet the

Table 1
The initial conditions of boost and glide phases.

Parameters	Boost phase	Glide phase
h (km)	6	60
V (m/s)	900	4000
γ (°)	0	0
θ_0 (°)	5	\
α_{max} (°)	\	10
$\alpha_{maxL/D}$ (°)	\	5

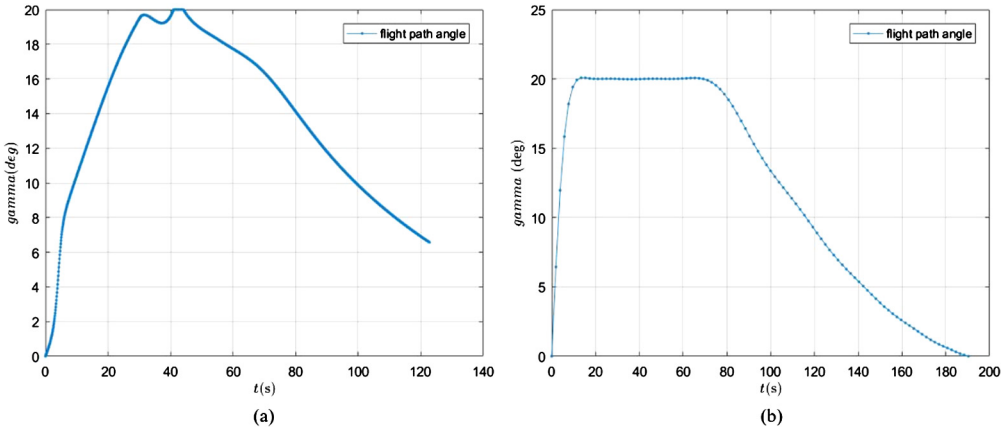


Fig. 7. The flight path angle profile of the boost phase. (a) Designed trajectory profile. (b) Optimized trajectory profile.

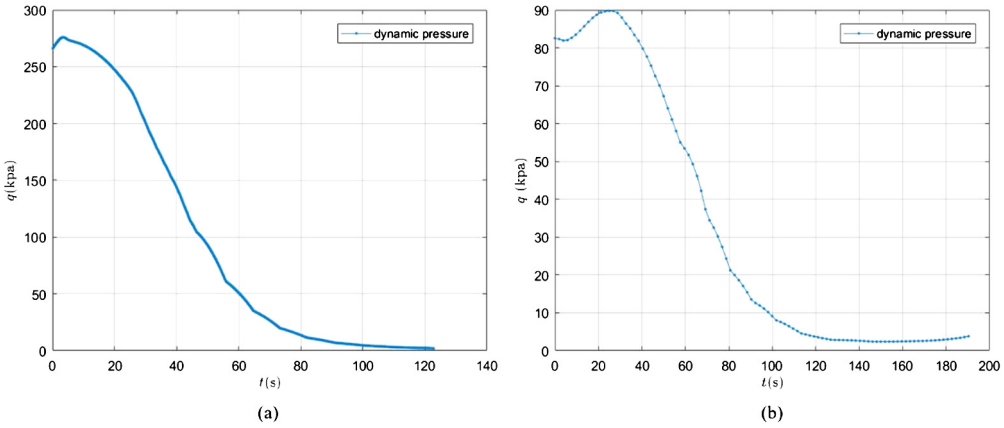


Fig. 8. The dynamic pressure profile of the boost phase. (a) Designed trajectory profile. (b) Optimized trajectory profile.

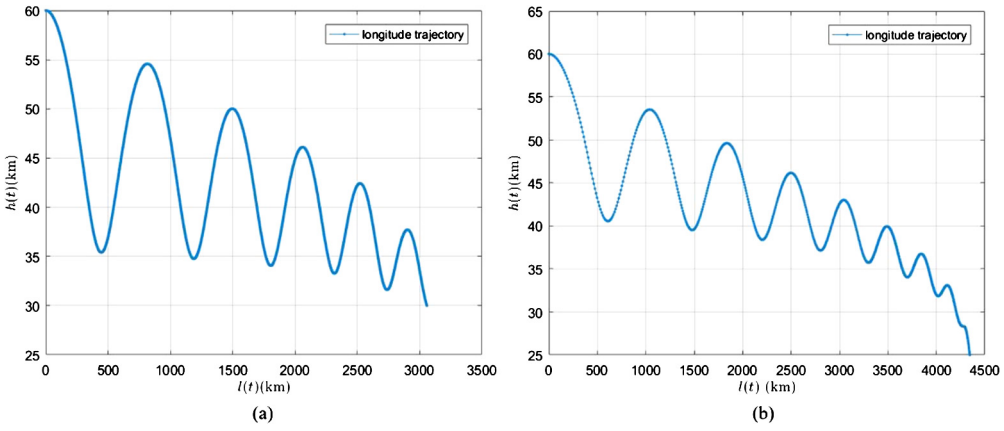


Fig. 9. The trajectory profile of the glide phase. (a) Designed trajectory profile. (b) Optimized trajectory profile.

requirement of 5000 m/s. In the first 45 s, the dynamic pressure (Fig. 8(a)) in the design part is larger than the limit, as a comparison, the optimization results show that the dynamic pressure (Fig. 8(b)) is always within the constraint range.

In the glide phase, the flight time is 1027 s and the flight range is 3058 km in design part (Fig. 9(a)). The optimization results indicate that the flight distance is longer and more maneuvering gliding (Fig. 9(b)) are taken. (See also Fig. 10.) The dynamic pressure profile (Fig. 11(a) and Fig. 11(b)) and heat flux density profile (Fig. 12(a) and Fig. 12(b)) are all within the constraints in the both design and optimization results. The results of the flight path angle shown in Fig. 13 corresponds to the results of the trajectory profile. When the HGV glides and maneuvers up and down outside the atmosphere, the change of its flight path angle is wave shape. The range of flight path angle in the optimization result (Fig. 13(b)) is smaller than that in the design result (Fig. 13(a)), which plays an important role in ensuring the stable flight of the vehicle.

According to the design optimization framework mentioned above, set the number of interval points in the optimization process as those [2,4] (low-density collocation) and [16,20] (high-density collocation). The results of the optimization process using low-density collocation points and high-density collocation points are shown in Fig. 14 respectively.

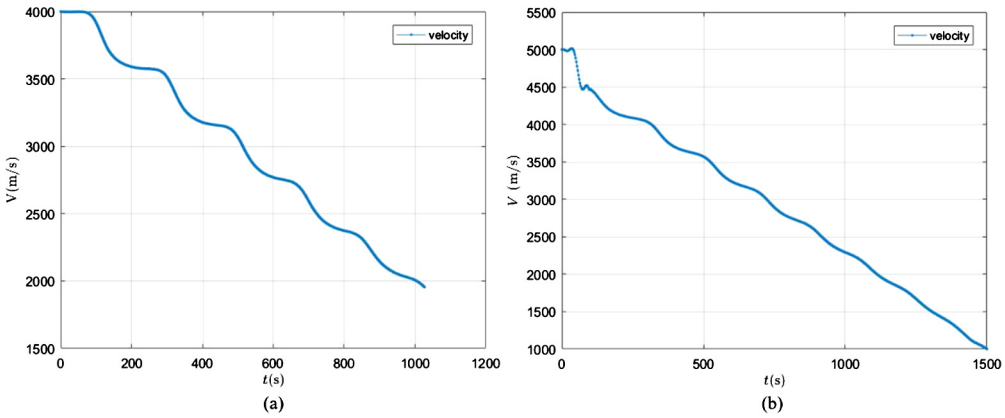


Fig. 10. The velocity profile of the glide phase. (a) Designed trajectory profile. (b) Optimized trajectory profile.

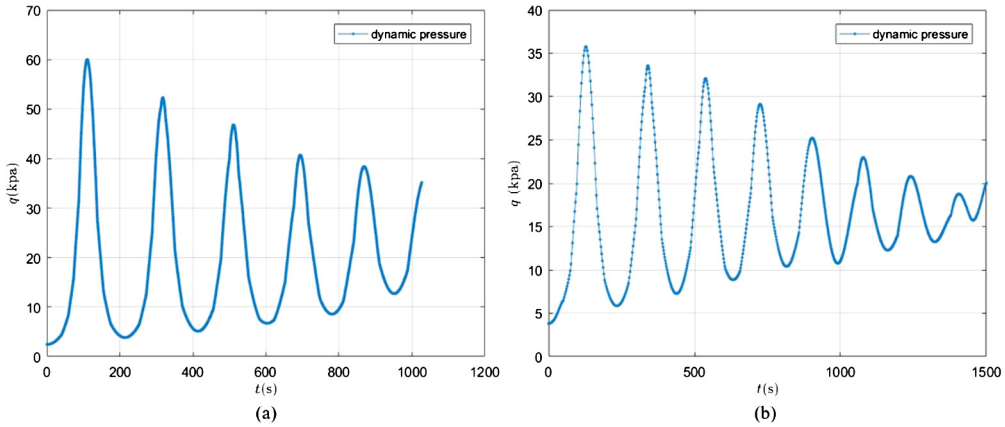


Fig. 11. The dynamic pressure profile of the glide phase. (a) Designed trajectory profile. (b) Optimized trajectory profile.

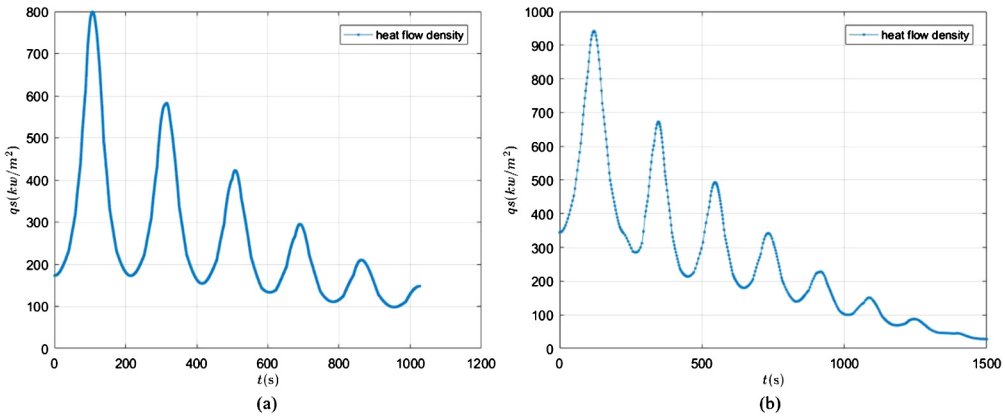


Fig. 12. The heat flux density profile of the glide phase. (a) Designed trajectory profile. (b) Optimized trajectory profile.

Table 2
The optimization parameters and relative constraints of boost and glide phases.

		Boost phase		Glide phase	
		Initial value	Terminal value	Initial value	Terminal value
State variants	V (m/s)	500	5000	5000	1000
	γ (°)	0	0	0	-5
	h (km)	6	60	60	25
	l (km)	$[0, 500]$		$[0, 5000]$	
Control variants	α (deg)	$[1, 12]$		$[9, 12]$	
Path constraints	q (kPa)	$q \leq 90$		$q \leq 90$	
	$\dot{q}_s$ (kw/m^2)	\		$\dot{q}_s \leq 1000$	

Taking the results of glide phase as an example, it can be seen from the figures that the optimization result of high-density collocation has a lower maneuver range than that of low-density collocation in gliding process. The reason is that it is necessary to re-adjust and optimize the trajectory based on the result of low-density collocation in order to reflect more flight information after increasing the collocation density. It can be found that the difference of flight distance between the both is small.

5.3. Simulation results with different initial conditions

In this part, the initial parameters of the glide phase are discussed. The initial parameters have an important influence on the trajectory profile and performance index, so it is necessary to dis-

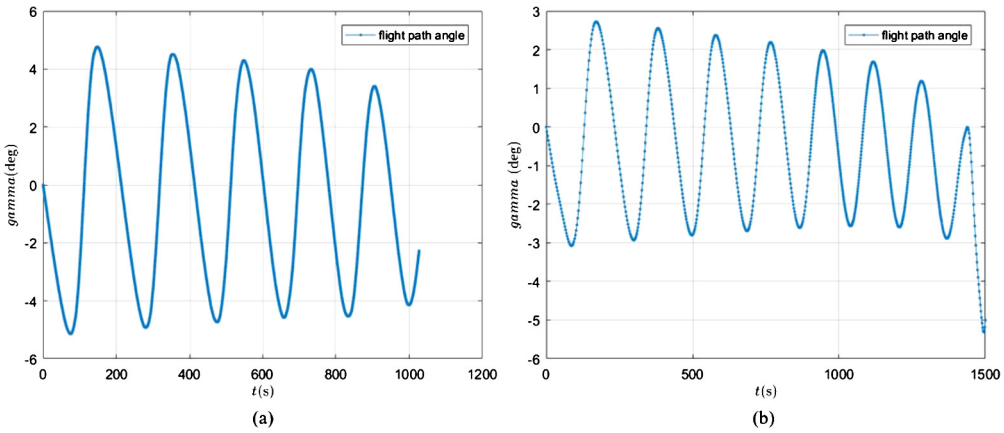


Fig. 13. The flight path angle profile of the glide phase. (a) Designed trajectory profile. (b) Optimized trajectory profile.

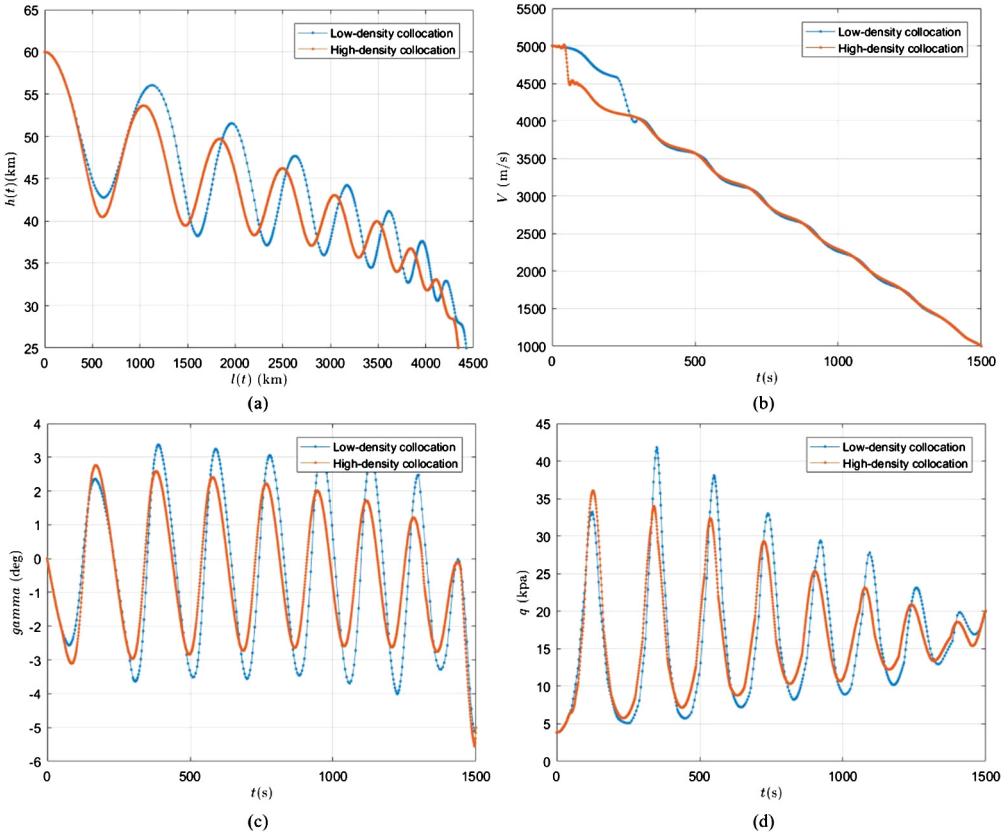


Fig. 14. The parameters comparison between the low-density collocation and high-density collocation in glide phase. (a) Trajectory comparison. (b) Velocity comparison. (c) Flight path angle comparison. (d) Dynamic comparison.

cuss this issue. Because the serious influence caused by initial flight path angle, velocity and altitude is more than other factors, the following analysis is focused on the three aspects.

5.3.1. Case 1: the influence of the initial flight path angle

Maintain the initial velocity and altitude value and modify the initial flight path angle to −2 deg, 0 deg, 1 deg, 2 deg, respectively. The specific parameters are shown in Table 3. As shown in Fig. 15, The influence of the initial flight path angle is mainly reflected in the maneuver amplitude during the gliding process and the flight distance characteristics (Fig. 15(a)). Changing the initial flight path angle can increase the distance and the maneuver amplitude. The relationship of height and Mach is shown in Fig. 15(b). The reason for modifying the flight path angle with these four values is that the range of flight path angle always varies in the range of [−3°, 3°] when the vehicle is gliding. The larger angle will increase the flight burden of the HGV. The results show that the dynamic pressure (Fig. 15(c)) and heat flux density (Fig. 15(d)) of the vehicle are kept within the constraint range, which indicates a good trajectory characteristics.

Table 3
Simulation initial conditions of the initial flight path angle.

	Initial flight path angle/(°)	Initial velocity/(m/s)	Initial altitude/(km)
Case 1	−2	5000	60
	0	5000	60
	1	5000	60
	2	5000	60

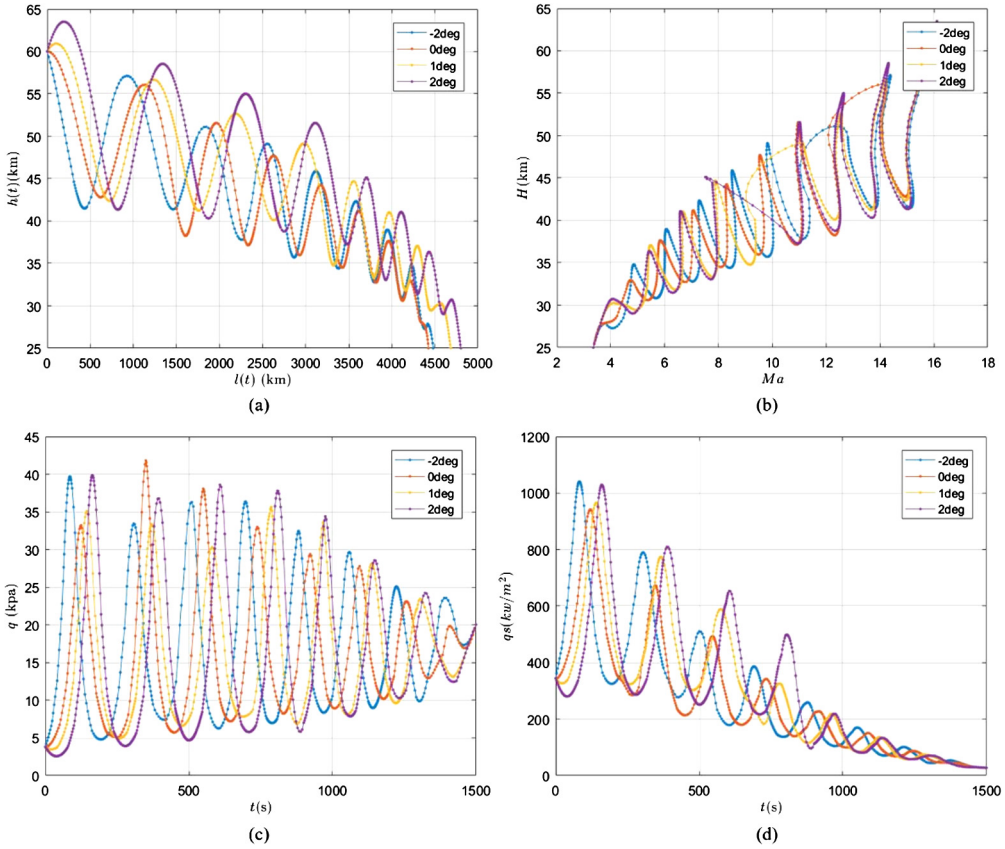


Fig. 15. The parameters comparison of different initial flight path angle in glide phase. (a) Trajectory comparison. (b) Altitude-Mach comparison. (c) Dynamic pressure comparison. (d) Heat flux density comparison.

Table 4
Simulation initial conditions of the initial velocity.

	Initial flight path angle/(°)	Initial velocity/(m/s)	Initial altitude/(km)
Case 2	0	4900	60
	0	5000	60
	0	5200	60
	0	5500	60

Table 5
Simulation initial conditions of the initial altitude.

	Initial flight path angle/(°)	Initial velocity/(m/s)	Initial altitude/km)
Case 3	0	5000	53
	0	5000	55
	0	5000	60
	0	5000	63

5.3.2. Case 2: the influence of the initial velocity

Different from Case 1, consider that set the initial velocity to be 4900 m/s, 5000 m/s, 5200 m/s, 5500 m/s respectively and keep the initial flight path angle and initial altitude. The specific parameters are shown in Table 4. As shown in Fig. 16, the effect of initial speed is mainly reflected in the flight distance characteristics. As can be seen from the trajectory profile shown in Fig. 16(a), different speed will increase the flight distance, but meanwhile, the heat flux density (Fig. 16(d)) will increase, but the parameter changes are within the constraints. Meanwhile, the dynamic pressure (Fig. 16(c)) also meets the constraints. The relationship of altitude and Mach is represented in Fig. 16(b), which has a better consistency in the trajectory profile.

5.3.3. Case 3: the influence of the initial altitude

Consider the initial altitude is 53 km, 55 km, 60 km, 63 km respectively. The simulation results are shown in Table 5.

Compared with the initial velocity and the initial flight path angle, different initial altitude has a greater impact on the flight distance. At first, the maneuver amplitude and the flight distance are increased (Fig. 17(a)), which greatly enlarge the dynamic pressure (Fig. 17(c)) and heat flux density (Fig. 17(d)) but always within the constraint range. These changes are also reflected in Fig. 17(b). In particularly, there is a special trajectory at 55 km, which has a small maneuver amplitude and a smooth trajectory because it is close to the quasi equilibrium glide condition. The other parameters are all meet the conditions.

6. Conclusions

A general trajectory design and optimization framework based on the pseudo-spectral method is proposed in this paper. Firstly, the trajectory is segmented according to the task scene, and the partial program angle control law is designed. Then, the optimization index is proposed for each phase and the LGR orthogonal collocation method is applied for the optimization process. During the optimization process, adjusting the density of the optimization collocation, and the results are compared in different stages. Finally, the control variable method is used to analyze the influences of different initial values on the gliding trajectory, further narrowing the range of the initial value option. The simulation results indicate the trajectories with high-density Gauss collocation have more flight information than those with low-density collocation and some trajectory parameters are different. In the related future work, the idea will focus on the adaptability of different aircrafts to the framework and the initial value estimation.

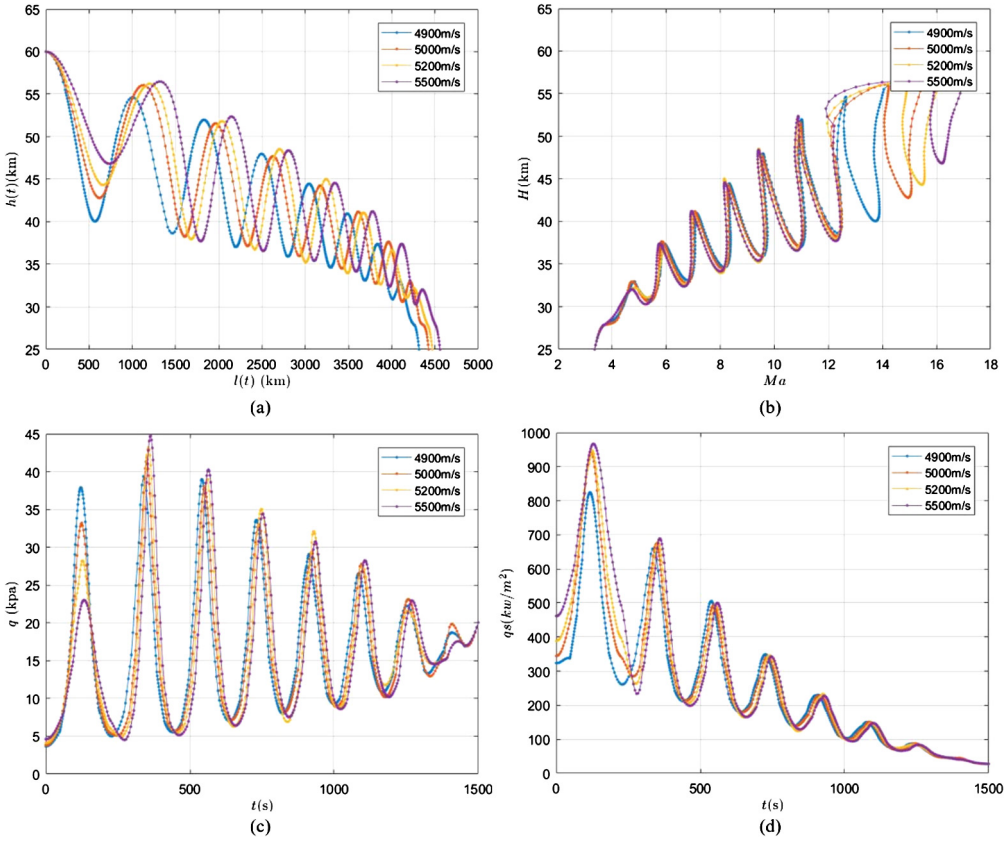


Fig. 16. The parameters comparison of different initial velocity in glide phase. (a) Trajectory comparison. (b) Altitude-Mach comparison. (c) Dynamic pressure comparison. (d) Heat flux density comparison.

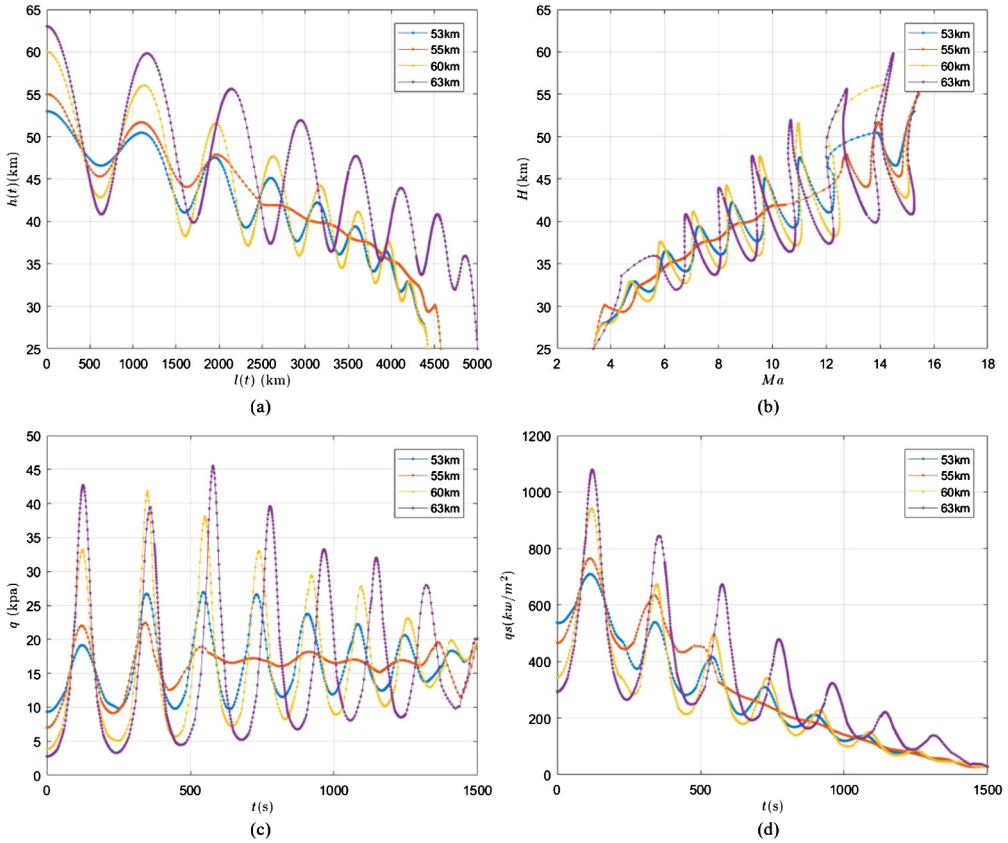


Fig. 17. The parameters comparison of different initial altitude in glide phase. (a) Trajectory comparison. (b) Altitude-Mach comparison. (c) Dynamic pressure comparison. (d) Heat flux density comparison.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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